5 **(a)** A straight rigid conducting wire of mass 0.14 kg and length 0.40 m is placed horizontally on a frictionless plastic plane. The plane is at an angle of 20° to the horizontal and is placed in a region of upwards uniform vertical magnetic field as shown in Fig. 5.1 and Fig. 5.2.

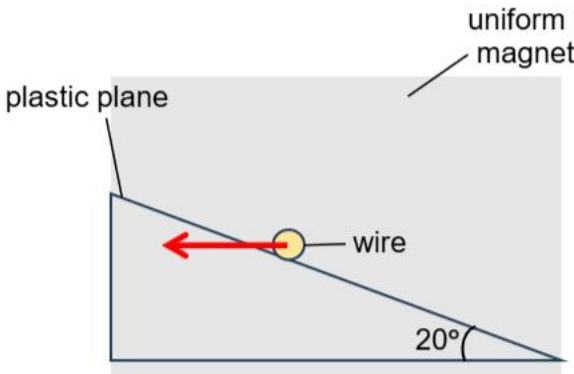


Fig. 5.1 (side view)

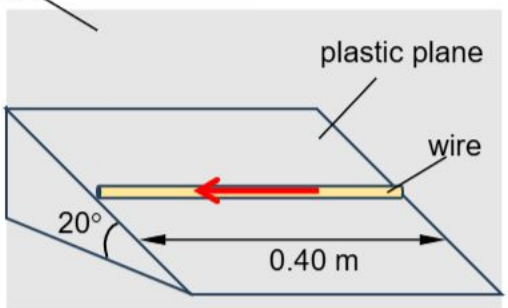


Fig. 5.2

To prevent the wire from sliding down the plane, a steady current I is passed through the wire so that it remains stationary on the plastic plane.

(i) On Fig. 5.1, draw an arrow to show the direction of the magnetic force acting on the wire. [1]

(ii) On Fig. 5.2, draw an arrow to show the direction of the current I in the wire. [1]

(iii) With reference to forces acting on the wire, explain why the wire remains stationary.

The direction of current I is perpendicular to the magnetic field/ flux density B and by Flemings left hand rule, experiences a force perpendicular to both I and B, acting leftwards.(Fig. 5.1)

[B1] The wire remains stationary when the component of the magnetic force upwards along the plane balances the component of weight downwards along the plane. [2]

[2]

(iv) Suggest and explain whether the wire will eventually slide up, remain stationary or slide down the plane if an alternating current of root-mean-square value of I is passed through the wire instead.

The current I will alternate between positive and negative values, and the component of magnetic force ($\propto I$) will alternate between acting upwards and downwards along the plane.

Since the average I and hence magnetic force between cycles is zero, it cannot balance the component of weight downwards along the plane. Hence the rod will slide down the plane.

Or?

The current I will alternate between positive and negative values, and the component of magnetic force ($\propto I$) will alternate between acting upwards and downwards along the plane. [2]

Since I_{rms}= 0.707 I_0, the time duration at which the a.c current causes a **net** upward force along plane (when upward $F_{m//}$ > W// to slope downwards) is shorter than when there is a net downward force. Hence the rod will slide down the plane.

(b) The current in **(a)** is disconnected, and the magnetic field is removed. The wire is now given a uniform net positive charge of $+2.0 \times 10^{-6}$ C.

(i) To keep the wire stationary in the same position on the frictionless plane, a uniform electric field of strength E is applied in the horizontal direction.

Determine the magnitude of E.

The electric force must be acting leftwards for wire to be stationary. $\sum F_{//} = 0$

$$F_{E//} = W_{//}$$

$$qE_{//} \cos\theta = mg \sin\theta \quad \text{[C1]}$$

$$E_{//} = \frac{mg}{q} \tan\theta$$

$$= \frac{(0.14)(9.81)}{2.0 \times 10^{-6}} \tan 20° \quad \text{[M1]}$$

$$= 2.5 \times 10^{5} \quad \text{[A1]}$$

E = ... N C^{-1} [3]

[Total: 9]